

DETECTING FISH HUNGER LEVELS BY MONITORING AND CLASSIFYING THEIR UNDERWATER SWIMMING PATTERNS

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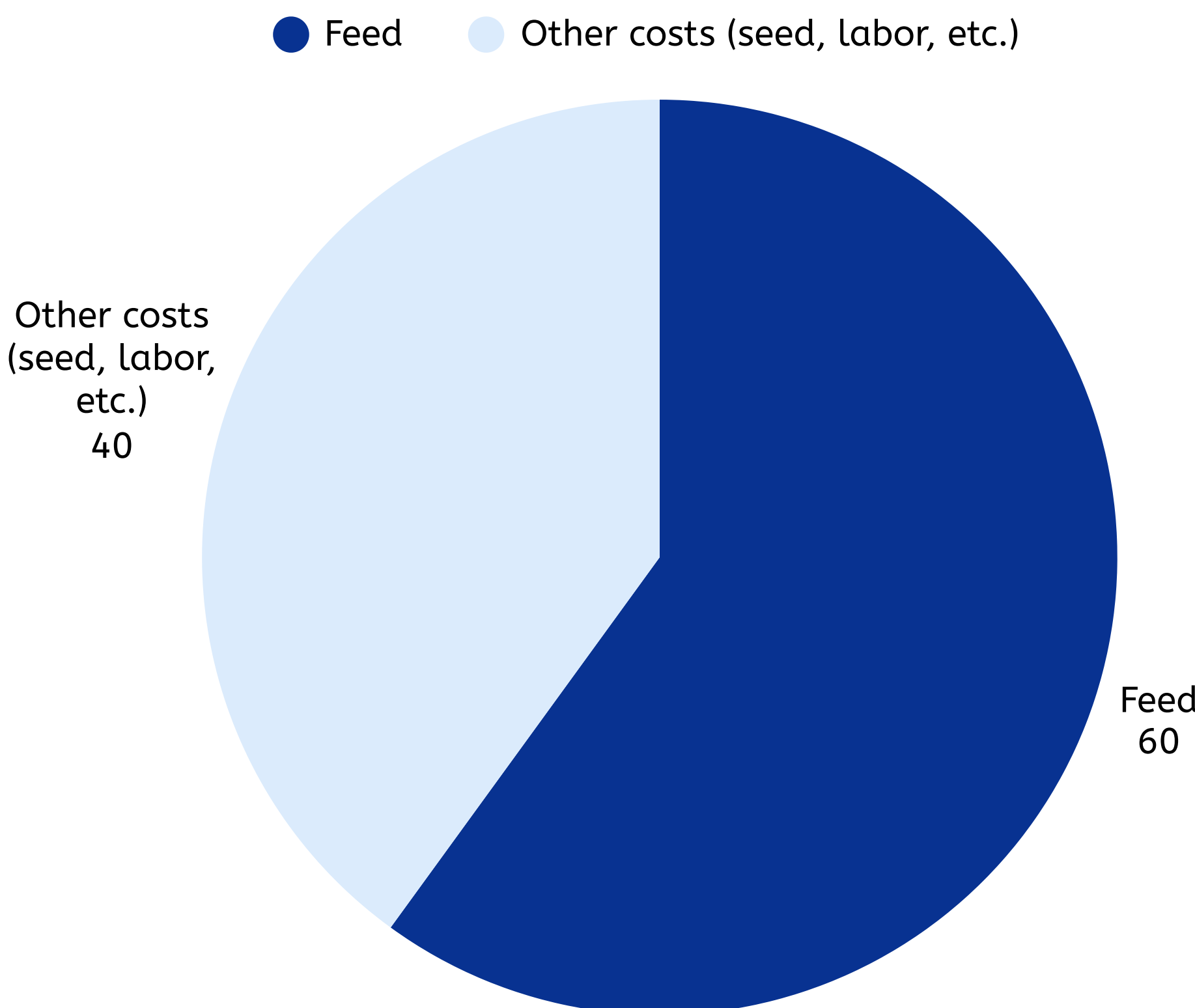
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Introduction

- 1 Fish farming is an important source of food and income in Bangladesh, especially for people with small ponds.
- 2 Feed is the biggest cost for many farmers, up to around 60% of production cost in some farms.
- 3 Most farmers feed at fixed times or just by looking at the pond surface.
- 4 Fish can change how they swim when they are hungry (speed, turning, depth, how close they group). A camera may be able to pick this up.

Production Cost



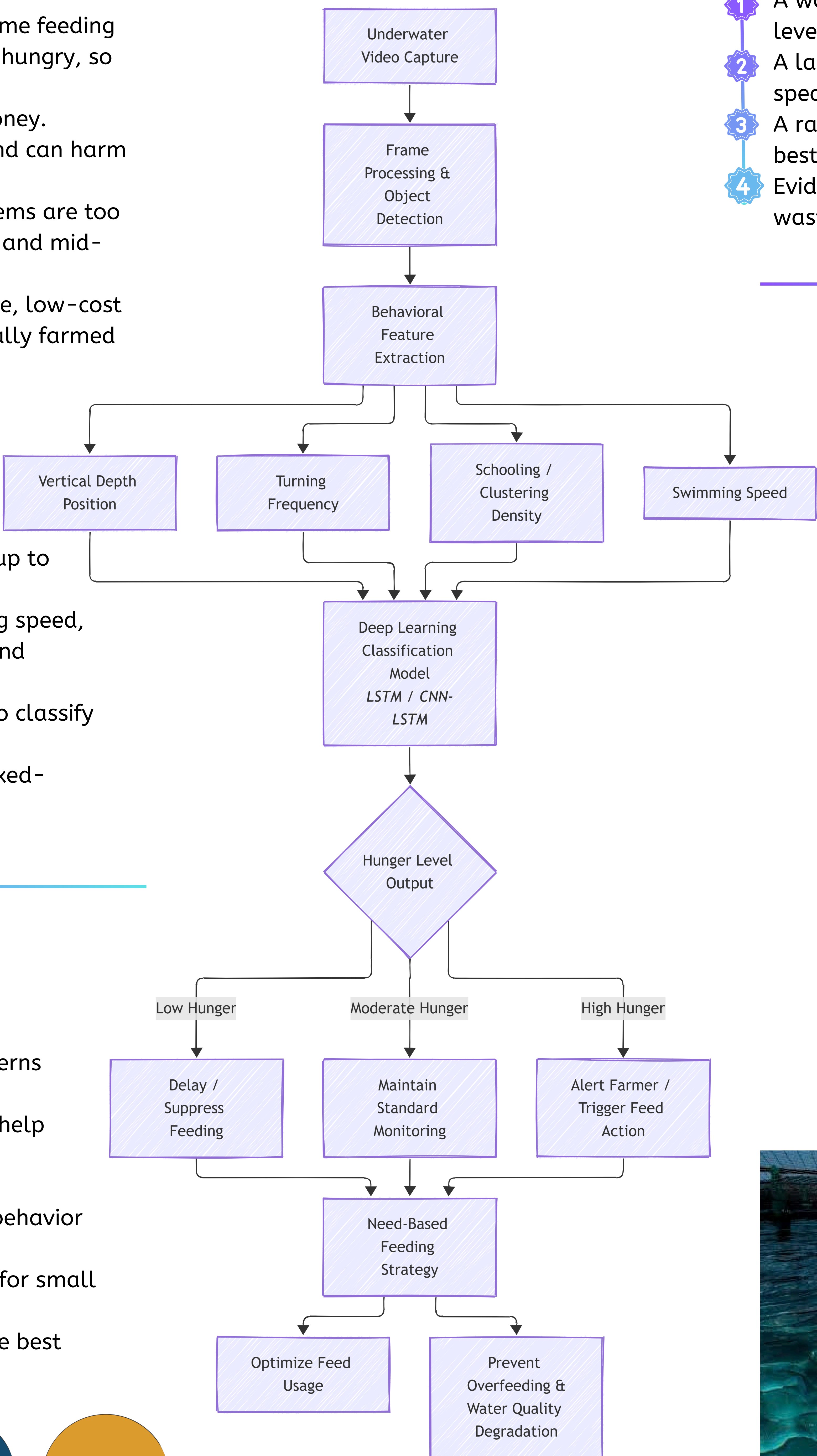
Methodology

- 1 **Record:** Film fish underwater at different times after feeding. Time since last feeding gives us the hunger label.
- 2 **Track:** Detect the fish and follow their paths frame by frame.
- 3 **Extract features:** Speed, turning, vertical position, schooling density.
- 4 **Train:** The model looks at short video clips and predicts Low, Moderate or High hunger.
- 5 **Test:** Try it on new videos the model hasn't seen, then compare need-based and fixed-schedule feeding (feed used, growth, water quality).

Problem Statement

- 1 **Inaccurate Feeding Methods:** Fixed-time feeding doesn't tell us if the fish are actually hungry, so overfeeding is common.
- 2 **Wasted Money:** Extra feed wastes money. Leftover feed also rots in the pond and can harm water quality.
- 3 **High-Cost Tech:** Modern feeding systems are too expensive and complicated for small and mid-sized farms.
- 4 **Lack of Local Tools:** There is no simple, low-cost way to estimate hunger level for locally farmed fish, and no dataset to build one on.

Methodology Architecture



Research Objectives

- 1 Build a low-cost, camera-based setup to record fish underwater.
- 2 Track the fish and measure swimming speed, turning frequency, vertical position and schooling density.
- 3 Train a model (LSTM or CNN-LSTM) to classify hunger as Low, Moderate or High.
- 4 Compare need-based feeding with fixed-schedule feeding.

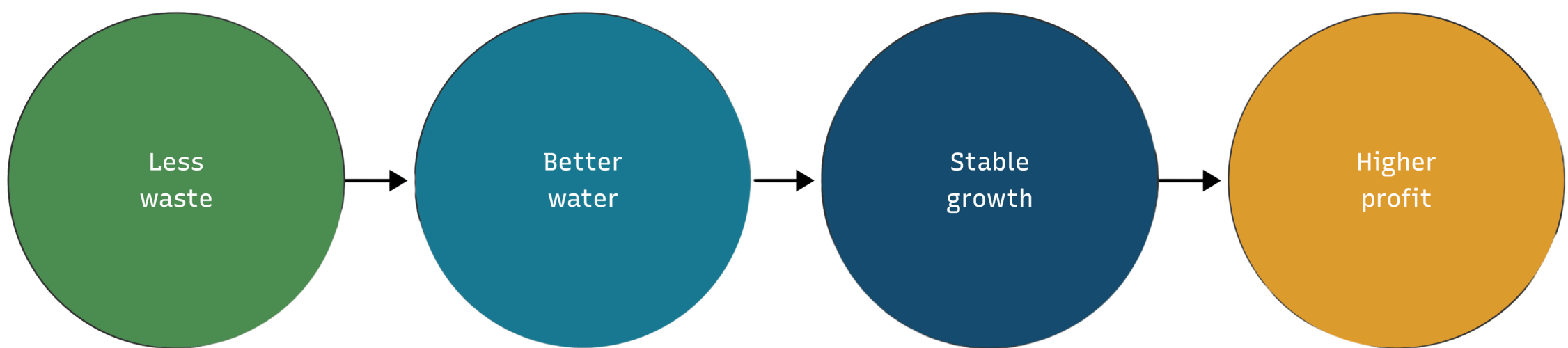
Research Questions & Contribution

RQ1: Can underwater swimming patterns show how hungry fish are?

RQ2: Can a low-cost camera system help farmers decide when to feed?

Our contribution:

- 1 A labeled dataset linking swimming behavior to hunger in local fish
- 2 A simple, real-time hunger indicator for small farms
- 3 Finding out which swimming cue is the best hunger signal



Expected Outcomes

- 1 A working prototype: underwater video in, hunger level out.
- 2 A labeled dataset, which is scarce for local fish species.
- 3 A ranking of which movement features signal hunger best.
- 4 Evidence on whether need-based feeding reduces wasted feed and helps water quality.

References

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